

Restyle Website Application Proposal: Managing a Wardrobe and Improving Fashion Sense



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Abstract

This project envisions the creation of a wardrobe management website that allows users to upload images of their clothing items, scan and categorize them, and visualize outfits on customizable 3D or 2D models. Unlike traditional wardrobe management tools, which are static and limited to basic organization, this platform will leverage image recognition and AI to provide dynamic categorization and outfit recommendations. The website will feature an intuitive drag-and-drop interface for designing outfits and will include functionality for personalized suggestions based on user preferences, such as season and formality. The expected outcomes include improved wardrobe organization, reduced decision fatigue, and increased creativity in outfit planning. The significance of this project lies in advancing wardrobe management tools by integrating modern technologies such as image recognition and 3D modeling to provide a seamless and engaging user experience.

1.0 Introduction

The wardrobe management website project is a web-based platform designed to simplify wardrobe organization and enhance outfit planning. By leveraging modern technologies like image recognition and 3D/2D modeling, the platform allows users to upload images of their clothing, scan and categorize them, and visualize outfit combinations on customizable models. Users often face challenges in organizing their wardrobe and deciding what to wear, leading to decision fatigue. This project aims to address these challenges by providing a user-friendly platform that combines wardrobe management tools with powerful visualization features.

1.1 Background

Wardrobe management can be a time-consuming and overwhelming task, especially for individuals with large or diverse clothing collections. Many people struggle with organizing their wardrobe, keeping track of what they own, and visualizing how different clothing items might look together. Traditional methods, such as manually sorting clothes or using static wardrobe apps, often fall short in addressing these issues. By introducing a platform that uses image recognition to categorize clothing items and 3D/2D modeling to visualize outfits, this project aims to revolutionize the way users interact with their wardrobe.

1.2 Motivation

The motivation for this project stems from the need for a tool that simplifies wardrobe management while fostering creativity in outfit planning. Users often forget what clothing items they own or find it difficult to visualize how certain items might pair together. This platform will reduce decision fatigue by categorizing clothing items automatically and providing outfit recommendations. Additionally, the use of 3D/2D modeling will allow users to experiment with different combinations, making the process of outfit planning more engaging and enjoyable.

1.3. Related Work

1.3.1 Using Image Recognition for Clothing Categorization

Applications like Google Lens and Pinterest have successfully implemented image recognition for object categorization. This project extends that concept to clothing items, enabling users to upload images of their wardrobe and have them automatically categorized by type, color, and formality.

1.3.2 Virtual Try-On Technology

Brands like Nike and Sephora have introduced virtual try-on features for shoes and makeup. Similarly, this project uses 3D/2D modeling to allow users to visualize clothing combinations on customizable models, enhancing the experience of outfit planning.

1.3.3 Wardrobe Management Apps

Applications like Cladwell and Stylebook focus on wardrobe organization but lack strong visualization features. This project bridges the gap by combining wardrobe management tools with 3D/2D visualization to provide a more interactive and engaging experience.

2.0 Problem Statement

Users often face challenges in organizing their wardrobe and visualizing outfit combinations. Existing solutions either focus on wardrobe management or virtual try-ons but fail to combine both seamlessly. Additionally, decision fatigue and lack of creativity in outfit planning are common issues. This project addresses these gaps by providing a platform that scans clothing items, categorizes them, and enables users to visualize outfits on customizable 3D/2D models.

3.0 Proposed Project & Significance

The proposed project is a wardrobe management website that combines image recognition, 3D/2D modeling, and outfit visualization. Users can upload images of their clothing, which are scanned and categorized automatically. The platform also allows users to drag and drop clothing items onto customizable models to create and visualize outfits. By integrating modern technologies, this project offers a unique solution to wardrobe management challenges, enabling users to organize their clothing more effectively and plan outfits creatively.

3.0.1 Scope

To keep development realistic and measurable, this project will define a clear scope and a “definition of done.” The minimum viable product (MVP) will include: user authentication, adding/editing/deleting clothing items with image upload, optional AI-assisted auto-filling of item fields (category/color), outfit generation using user-selected filters (season and formality), saving generated outfits, and previewing an outfit on a basic 3D mannequin. Features that are explicitly out of scope for the initial release include full-body scanning, physics-based cloth simulation, highly accurate virtual try-on, and social/sharing features. The project will be considered complete when the MVP features work end-to-end for a typical user flow (sign in → add items → generate outfits → save → preview) with stable performance and documented installation steps.

3.0.2 Success Criteria (Definition of Done)

This project will be considered successful when a user can complete the full workflow reliably: create an account, log in, add clothing items (with or without image scanning), generate outfits using filters (season and formality), save selected outfits, and preview an outfit in the preview page. The scanning feature will be considered complete when it returns a predicted category and color for an uploaded image and the user can confirm or edit the results before saving. The outfit generator will be considered complete when it produces multiple outfit combinations from the user’s wardrobe and respects selected filters. Finally, the project deliverables will include a working deployment-ready build, a short user guide, and installation instructions that allow another student to run the system locally.

4.0 Objectives (Step-by-step requirements and UX)

4.1 Overall Project Objective

To develop a wardrobe management website that scans uploaded clothing images, categorizes them, and enables users to visualize outfits on customizable 3D/2D models.

4.2 UX Design and Functions

4.2.1 Getting Started with the Wardrobe Website

The user will visit the wardrobe website and create an account by providing basic information such as their email, username, and password. Upon logging in, the user will be greeted by a homepage that introduces the platform's features, such as uploading clothing items, creating outfits, and visualizing them on models.

4.2.2 User Creating a Profile

Users will have the option to create a personal profile, where they can input their body dimensions (e.g., height, weight) to generate a customizable 3D model. Alternatively, users can upload a photo of themselves to create a 2D model that reflects their appearance.

4.2.3 Uploading and Scanning Clothing Items

Users can upload images of their clothing items, such as tops, bottoms, and shoes. The system will use image recognition to scan the images, remove the background, and categorize the items based on attributes like type, color, and formality. Users will have the option to edit or confirm the categorization

4.2.4 Using the 3D/2D Model for Outfit Design

Users can drag and drop clothing items onto their customizable 3D/2D model to visualize outfit combinations. The platform will also provide outfit recommendations based on user preferences, such as season, formality, and color compatibility.

5.0 Activities

5.1 Functionality

The platform will scan and categorize clothing items using image recognition, enabling users to manage their wardrobe efficiently. Users can visualize outfits on customizable models and receive personalized outfit recommendations.

5.2 User-Friendly Design

The platform will feature an intuitive drag-and-drop interface, making it easy for users to design outfits. Clear

instructions and a visually appealing design will ensure a seamless user experience.

5.3 Security

The platform will implement standard security practices, including encrypted communication (HTTPS), password hashing (bcrypt), and secure data storage, to protect user information.

5.4 Privacy Details

In addition to password hashing, the application will apply practical security controls for authentication, file uploads, and user data privacy. JWTs will be issued with an expiration time and validated on protected routes to ensure only authenticated users can access wardrobe and outfit data. The backend will validate file types and size limits for image uploads to reduce risk from malicious files, and it will restrict image access so users can only view their own uploaded items. The system will also support deletion behavior that removes a clothing item record and its associated image reference when the user deletes an item. These measures help protect user accounts, reduce abuse of the upload endpoints, and ensure wardrobe images are treated as private user data.

6.0 Development Environment

6.1 Software Requirements

Software	Purpose
React.js	Frontend framework
Node.js + Express	Backend API
PostgreSQL	Database for storing user data
Python + OpenCV	Image processing
Three.js	3D model rendering

6.2 Hardware Requirements

Hardware	Purpose
Laptop/PC	Development and testing

6.3 Architecture Overview

The system will use a modular web architecture with a React frontend, a Node/Express backend API, a PostgreSQL database, and a separate Python FastAPI service for image-based clothing prediction. The React client handles routing and UI pages (wardrobe, generator, preview) and communicates with the Express API via JSON requests. The Express backend manages authentication (JWT), business logic (wardrobe CRUD, outfit generation), and database access (PostgreSQL). When a user uploads an image and requests scanning, the Express server forwards the image to the FastAPI prediction service, which performs preprocessing and returns predicted attributes (such as clothing category and dominant color). This separation keeps the ML component independent, allowing the main app to function even if the prediction service is unavailable.

7.0 Reports and Products

The final product will be a wardrobe management website that allows users to upload and categorize clothing items, visualize outfits on 3D/2D models, and receive personalized outfit suggestions. Additional deliverables include the source code and a detailed user guide.

8.0 Schedule

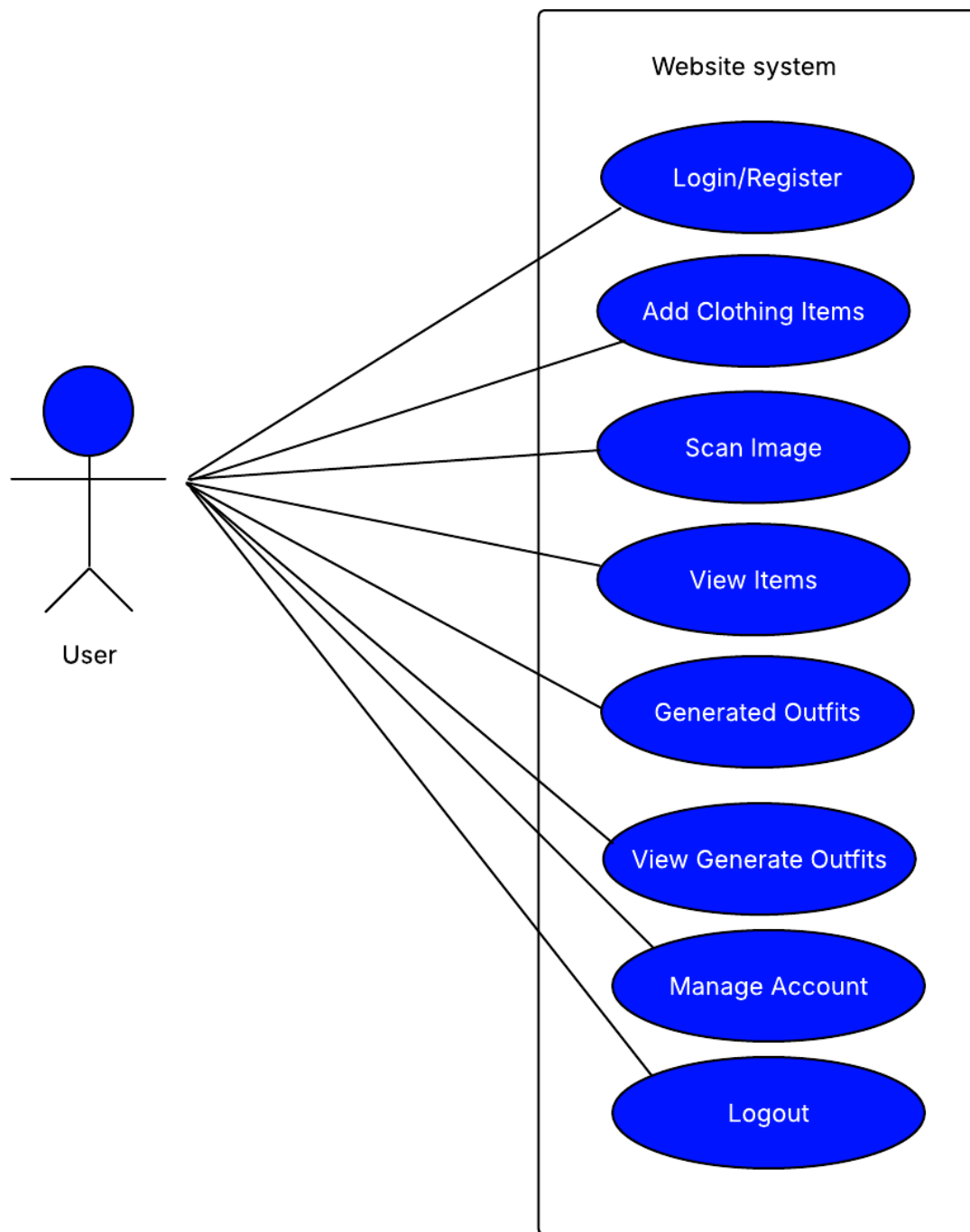
The schedule is organized by milestones so the team completes an end-to-end MVP early and reduces risk. The first phase focuses on requirements, database schema, and authentication. The second phase implements wardrobe CRUD and image upload, followed by outfit generation and saving outfits. The final phase integrates the PyTorch/FastAPI scanning service, completes the 3D preview, and finishes testing, documentation, and deployment preparation.

Task	Duration	Status
Research and Planning	1 week	Pending
Frontend Development	1 week	Pending
Backend Development	1 week	Pending
Image Processing Integration	1 week	Pending
3D/2D Model Implementation	1 week	Pending

Testing and Debugging	1 week	Pending
Final Deployment	1 week	Pending

9.0 UML Diagrams

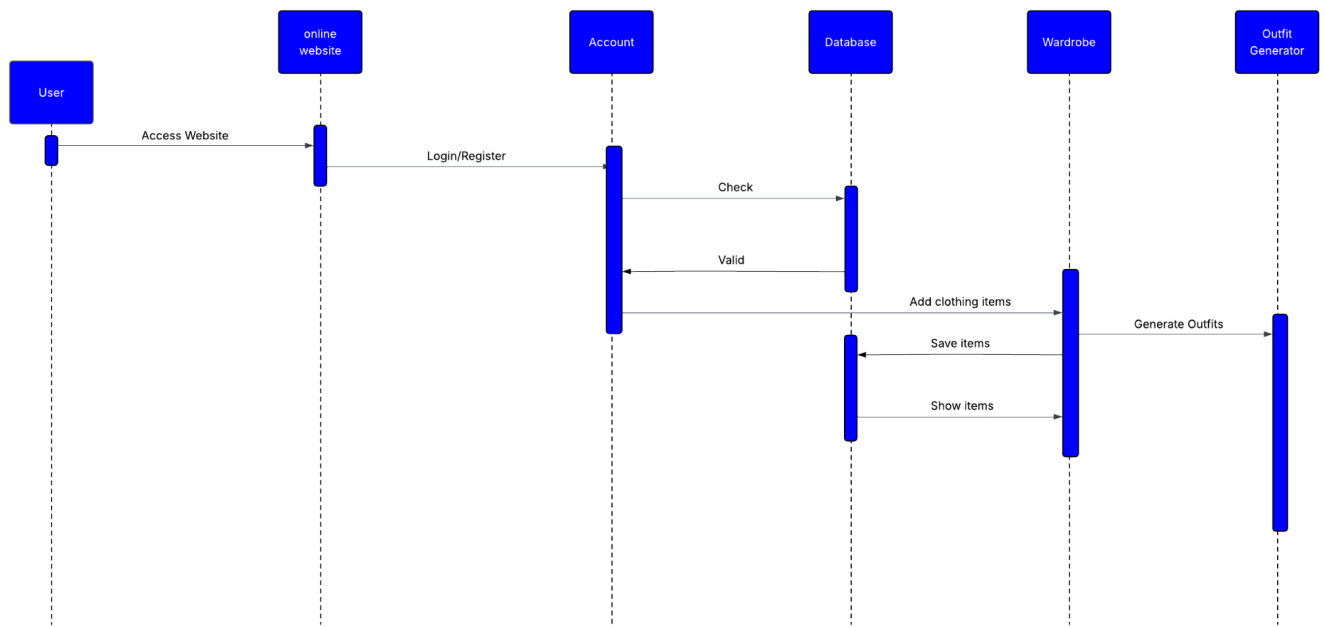
9.1 Use Case Diagram



9.2 Sequence Diagram

The sequence diagram starts on the user accessing the website. Users can access the website without logging in but they are very limited to what they can do. Once logged in, the users can add clothing items in the wardrobe. The items will be saved in the database then it will show the saved items in the wardrobe. With

various clothing items, the user can now generate outfits.



9.3 Data Model (Database Schema Overview)

The database will store user accounts, clothing items, and saved outfits. A users table will store user identity information (email/username) and a securely hashed password. A clothing_items table will store each item's user ownership and attributes such as name, category, color, season, formality, and an image path/URL. A outfits table will store saved outfits for each user along with metadata such as the filters used during generation (season/formality) and a timestamp. Because each outfit contains multiple clothing items, an outfit_items join table will map outfits to clothing items and optionally store a slot label (top, bottom, shoes) to keep outfit structure consistent. This schema supports efficient queries such as “all items for a user,” “all outfits for a user,” and “all items in an outfit.”

10.0 Implementation

Opening the website greets the user in the homepage showing different kinds of clothing with descriptions on each item. It also shows descriptions of each page and steps on how to use each page in the website. The users are very limited in what they can do as guests so logging into the website is required.

10.1 Sign-In

Clicking on the login/signup button, sends the user to the login page. If they already have an account, they can input their email address and password. Otherwise, they click on the sign up button below to create an account.

The image shows a login interface for a website called 'Restyle'. At the top, the word 'Restyle' is centered in a dark font. Below it, a light beige rectangular area contains a white login card with rounded corners. The card has a subtle orange glow. Inside the card, the heading 'Welcome Back' is in a bold, italicized serif font. Below the heading is the text 'Sign in to access your wardrobe' in a small, grey sans-serif font. There are two input fields: the first is labeled 'EMAIL ADDRESS' in small grey text above it, and contains the placeholder 'Enter your email'; the second is labeled 'PASSWORD' in small grey text above it, and contains the placeholder 'Enter your password'. Below these fields is a dark grey button with the text 'Sign In' in white. At the bottom of the card, there is a line of text: 'Don't have an account? [Sign Up](#)', where 'Sign Up' is a red, underlined link.

10.2 Sign-Up

If the user clicks on the “Sign up”, they are sent to the sign-up page where they need to enter their username, email address and password. The user can choose any username and a password that contains at least 6 characters.

Restyle

Create Account

Start optimizing your wardrobe today

USERNAME

EMAIL ADDRESS

PASSWORD

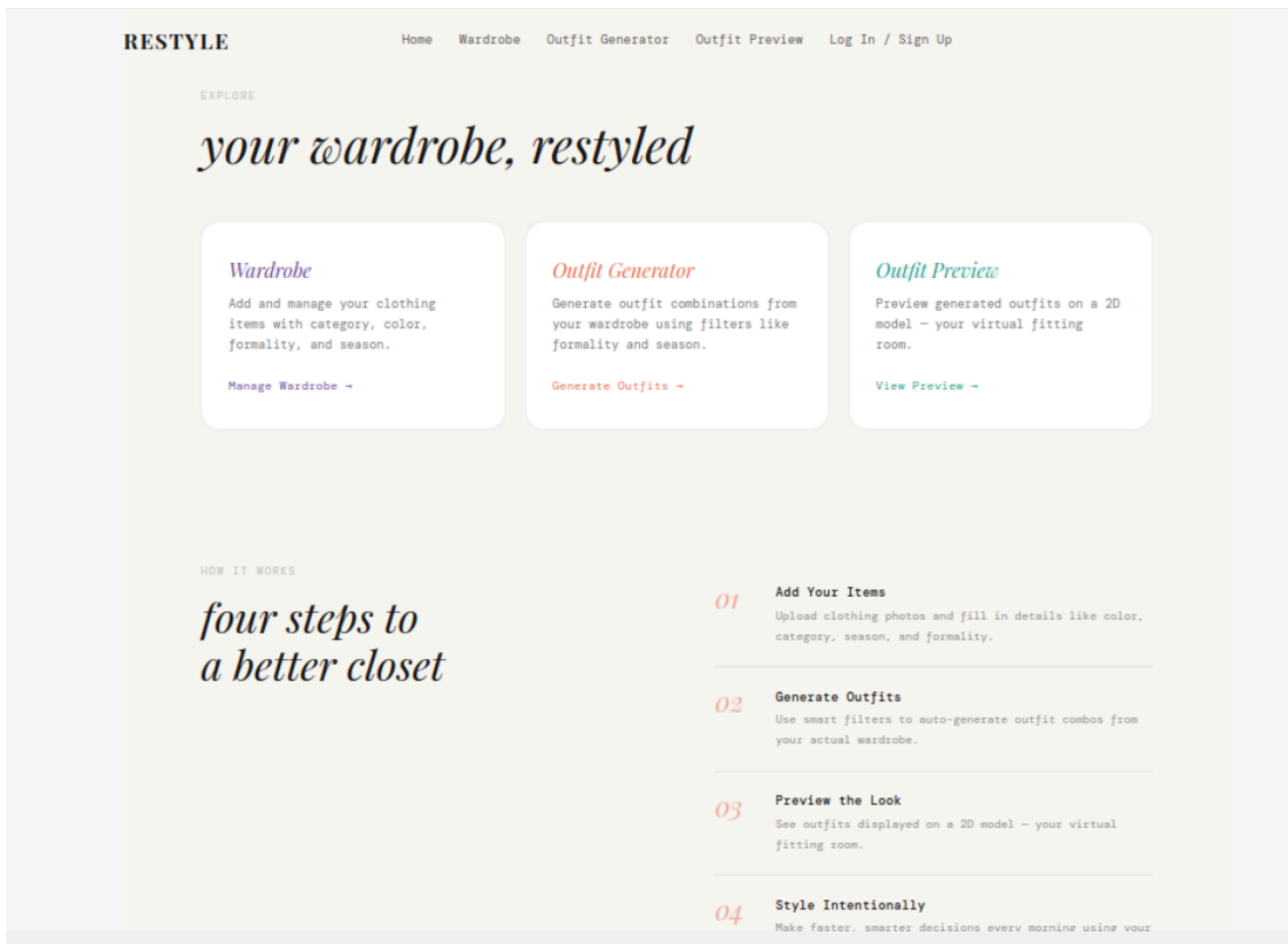
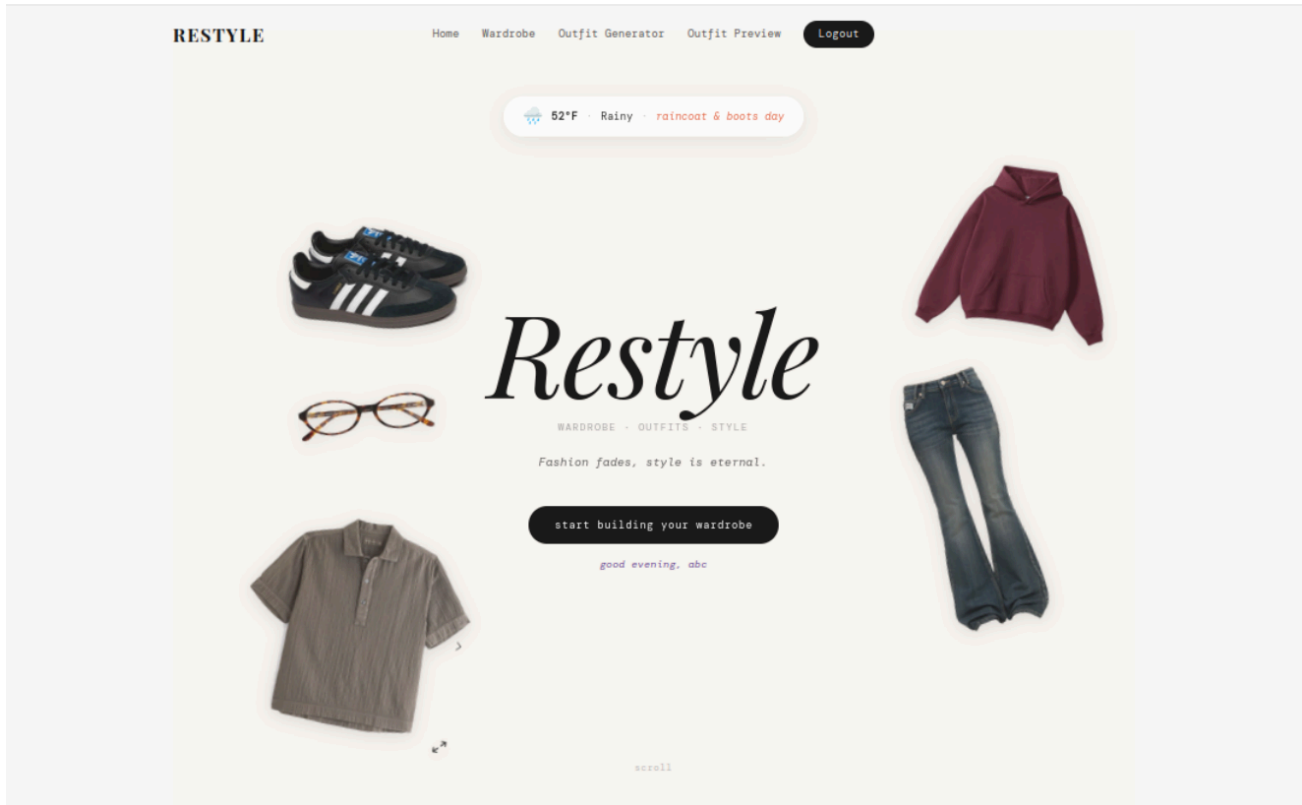
CONFIRM PASSWORD

Create Account

Already have an account? [Sign In](#)

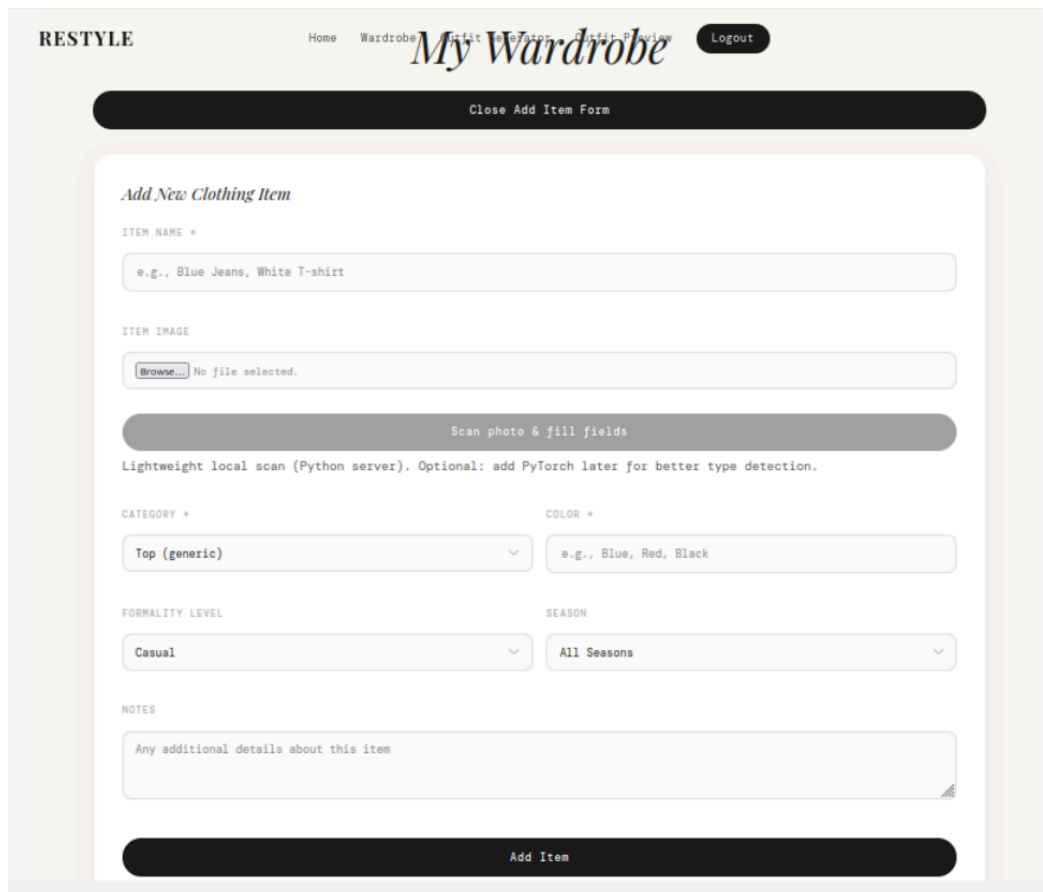
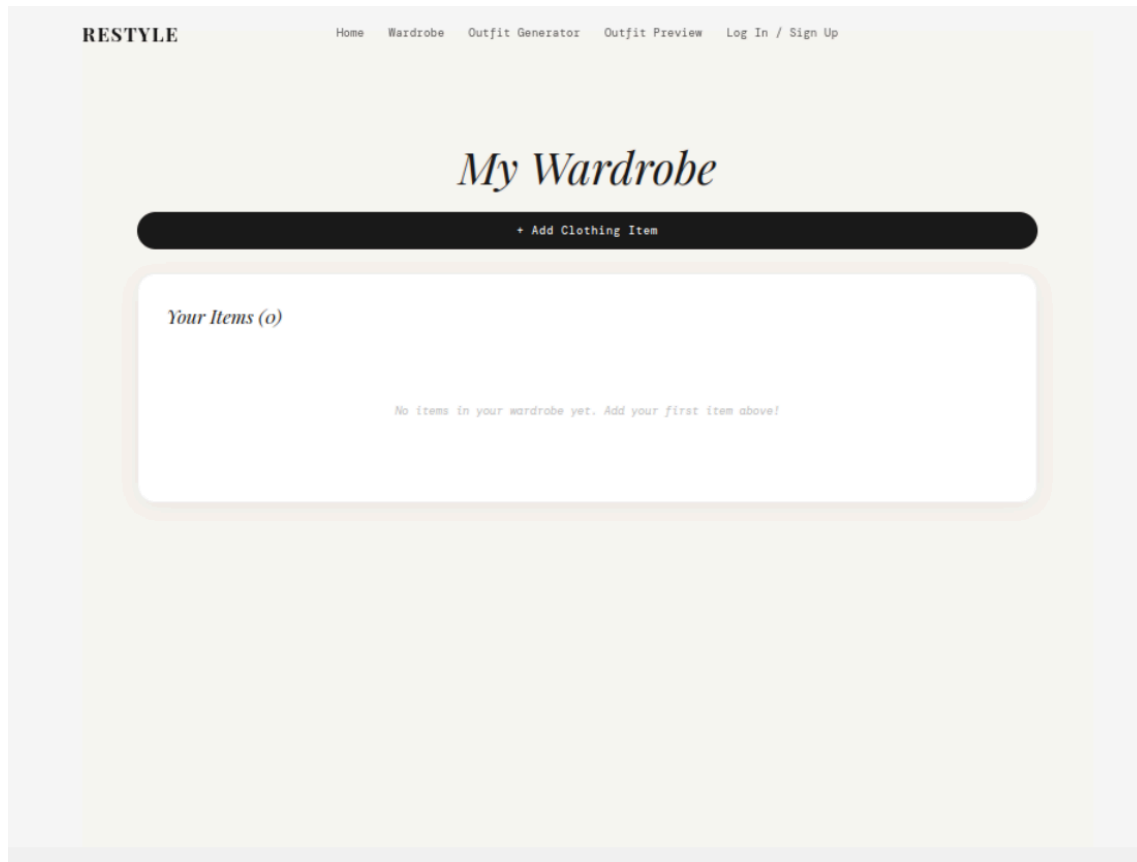
10.3 Homepage

Once the user is logged in, they are sent back to the homepage. The user is greeted with good morning, afternoon and evening based on the current time. The user can point their mouse to the different items in the list and it shows them a description of the item. Scrolling down the homepage, it will show the description of each page and clicking on the arrows will navigate them to the page they chose. There are also steps the users can follow in order to have a better experience.



10.4 Wardrobe

The wardrobe page is where the user will input the clothing and add the item in the wardrobe. The user needs to add clothing in their wardrobe to generate different outfit combinations in the Outfit Generator page. To add clothing items the users are required to input the item name, category, and color of the clothing. The user can also input an image of the clothing they chose and scan using PyTorch. When the user clicks the scan photo & fill fields, it will auto fill out each field. Identifying what kind of clothing with scan is not always accurate so the users are able to change each field in case the scan gives the wrong information. After filling out each field, they can add items where the user can see the added items in “Your Items”.



RESTYLE

[Home](#)
[Wardrobe](#)
[Outfit Generator](#)
[Outfit Preview](#)

Logout

Close Add Item Form

Add New Clothing Item

ITEM NAME *

ITEM IMAGE

Scan photo & fill fields

Lightweight local scan (Python server). Optional: add PyTorch later for better type detection.
 This is a summer-time black Shirt. (model confidence 80%) – edit below if anything looks wrong.

CATEGORY *

Shirt

▼

COLOR *

FORMALITY LEVEL

Casual

▼

SEASON

Summer

▼

NOTES

Any additional details about this item

Add Item

RESTYLE

[Home](#)
[Wardrobe](#)
[Outfit Generator](#)
[Outfit Preview](#)

Logout

Your Items (9)

Black T-shirt

×



Category: t-shirt
 Color: **black**
 Formality: casual
 Season: summer

watch

×

Category: accessory
 Color: **black**
 Formality: formal
 Season: all

watch

×

Category: accessory
 Color: **white**
 Formality: casual
 Season: all

shoes

×

Category: shoes
 Color: **white**
 Formality: casual
 Season: all

shoes

×

Category: shoes
 Color: **black**
 Formality: casual
 Season: all

jeans

×



Category: bottom
 Color: **Blue**
 Formality: casual
 Season: all

10.4.1 ML & PyTorch Scanning

The “Scan photo & fill fields” feature will be implemented through a separate prediction service that uses

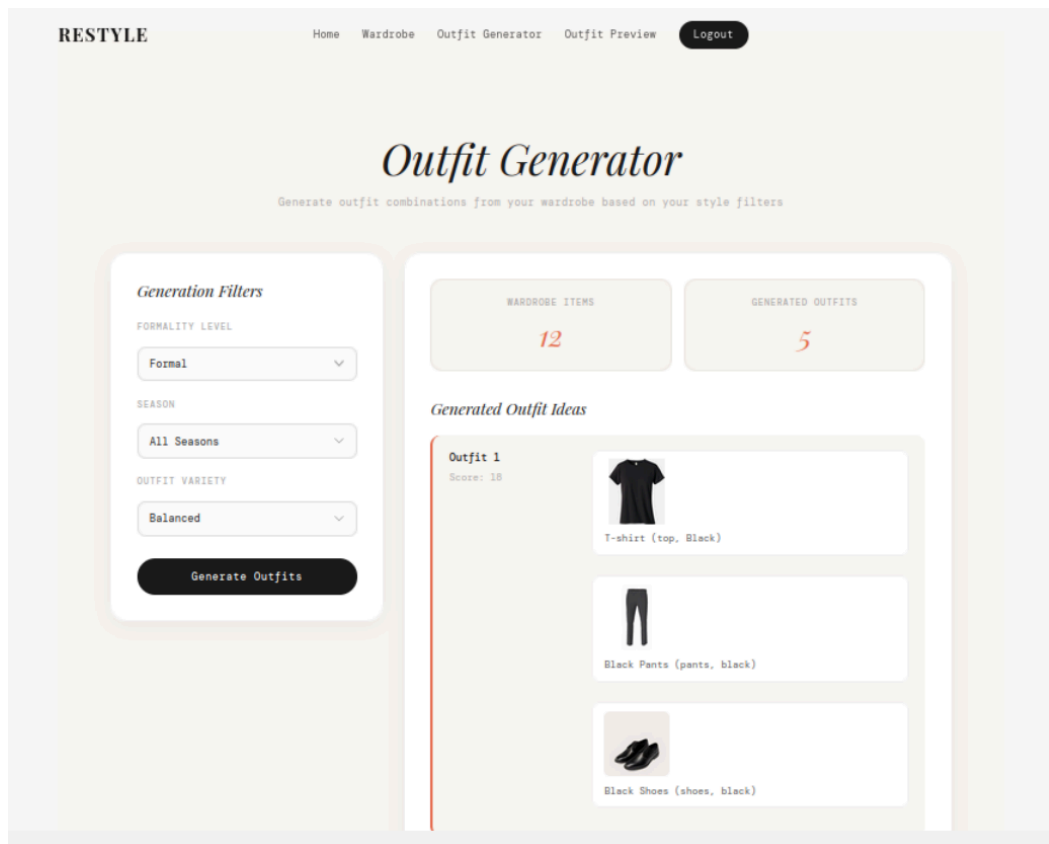
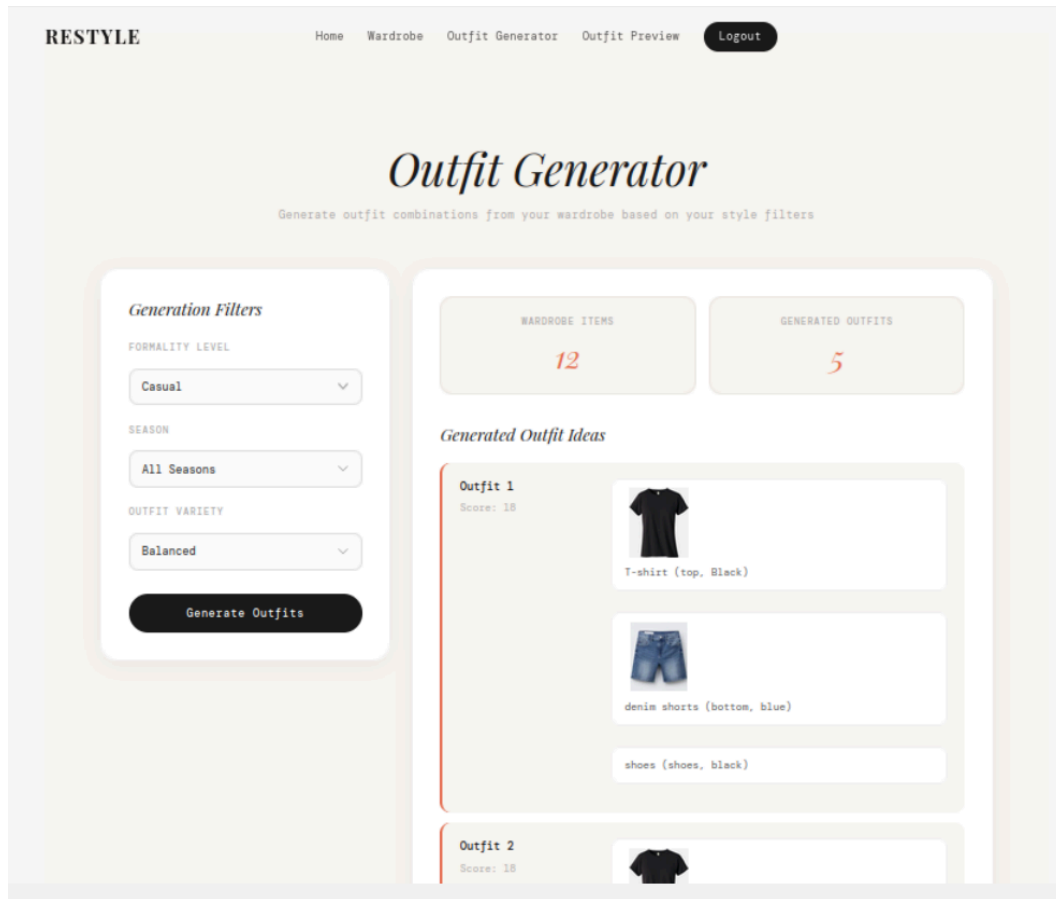
PyTorch-based models to estimate clothing attributes from an uploaded image. After the user selects an image, the backend sends it to the FastAPI service, which performs standard preprocessing (resize/normalize and optional background cleanup) and returns predictions such as clothing category (e.g., shirt, pants, shoes) and a dominant color label. Because real-world clothing photos vary widely, the scan results are treated as suggestions: the UI will always allow the user to confirm or edit the fields before saving. To evaluate the scanning feature, the team will test the predictor on a small labeled set of clothing images and report basic accuracy (e.g., category top-1 accuracy) and common failure cases, ensuring the feature is helpful even when it is not perfect.

10.5 Outfit Generator

The Outfit Generator page, as the title of the page suggests, generates outfits using beam search algorithms, selection heuristics and lightweight scoring. Before generating outfits, the users can use the generation filters that will improve on how they want their outfits to be generated. The first two filters: formality level and season. These are based on what the users filled out for their item in the wardrobe page. If the clothing item's formality level and seasons are the same as the generation filters, it will most likely show

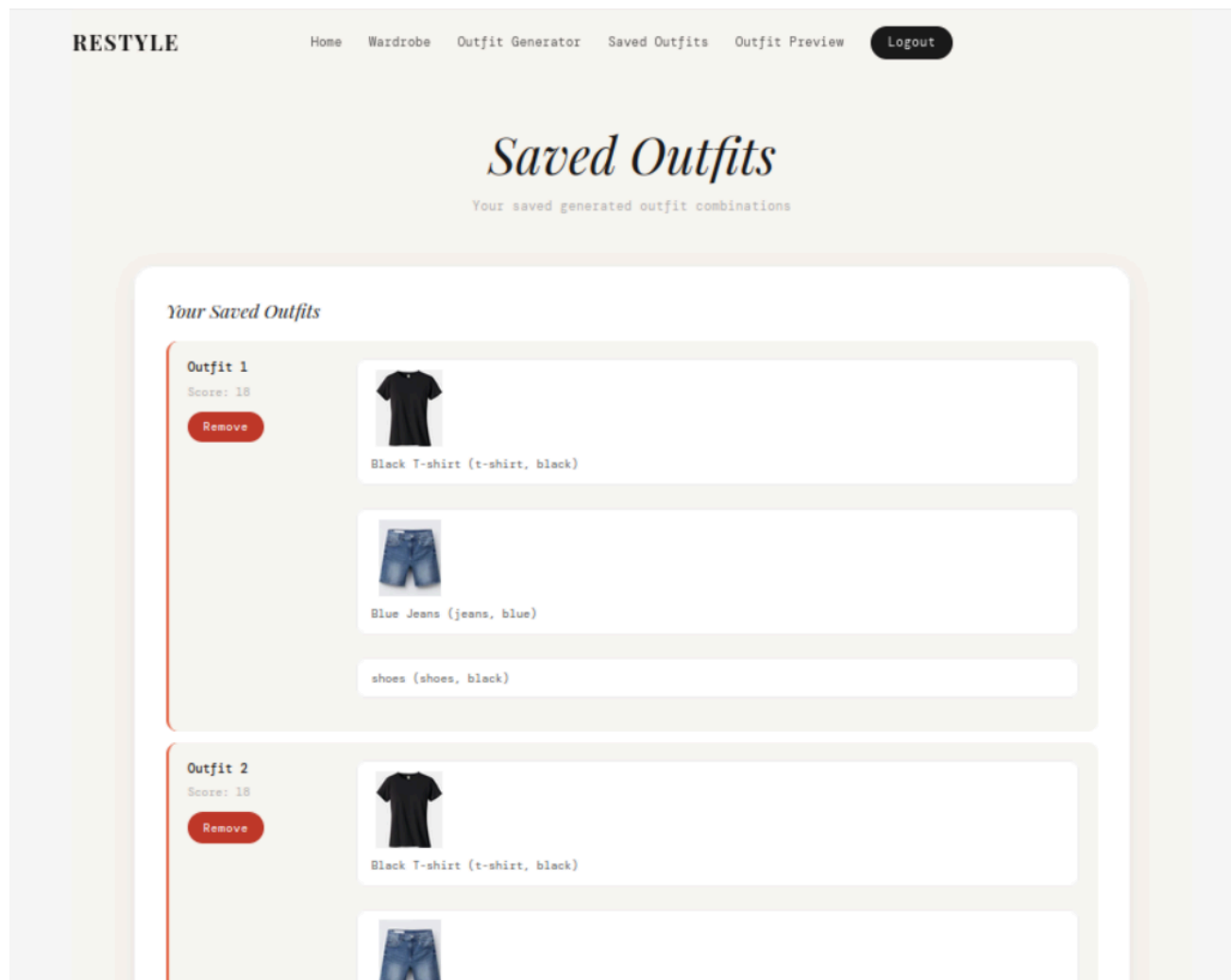
that item in the outfit generator. The third filter is outfit variety. There are three options: focused, balanced and more variety. Focused generates three outfits, balanced generates five outfits and more variety generates eight outfits. To generate outfits, the system will treat outfit creation as a constrained selection problem: it chooses compatible items across required slots (for example, top + bottom + shoes) while applying the user's filters. "Beam search" will be used to keep only the best-scoring partial outfit combinations at each step instead of trying every possible combination. The scoring will prioritize matching the selected season and formality, avoid repeating the same items too often, and encourage variety when the user selects the "more variety" option. This approach balances speed and quality so the generator can produce multiple outfits quickly even when the wardrobe grows.

The screenshot shows the RESTYLE Outfit Generator web interface. At the top, a navigation bar includes links for Home, Wardrobe, Outfit Generator, Outfit Preview, and a Logout button. The main heading is "Outfit Generator" with a subtitle "Generate outfit combinations from your wardrobe based on your style filters". On the left, a "Generation Filters" panel contains three dropdown menus: "FORMALITY LEVEL" set to "All Levels", "SEASON" set to "All Seasons", and "OUTFIT VARIETY" set to "Balanced". A "Generate Outfits" button is at the bottom of this panel. On the right, a large white box contains the instruction: "Select your preferences and click 'Generate Outfits' to get started. Your wardrobe items will be used to build outfit combinations."



10.6 Saved Outfits

The Saved Outfits feature allows users to store outfit combinations they like for later viewing and reuse. After outfits are generated, the user can select an outfit to save, which creates a saved outfit record in the database and links the chosen clothing items to that outfit. Saved outfits will be displayed in a list/grid view that shows the outfit's items and basic metadata such as the season/formality filters used. Users will also be able to delete saved outfits, which removes the saved outfit record and its item links without deleting the underlying clothing items from the wardrobe.



10.7 Outfit Preview

The Outfit Preview page provides a visual representation of a selected outfit on a mannequin model. The initial implementation will focus on a stable preview experience: loading a mannequin model and displaying the selected clothing items in a consistent layout, with camera controls to rotate and zoom. If full 3D clothing

fitting is not feasible within the project timeline, the preview will still provide value by presenting the outfit as layered textures or positioned clothing images on the mannequin. This page supports the project goal of helping users visualize combinations before choosing what to wear.

11.0 Testing and Integration

Testing will include unit testing of backend endpoints (authentication, wardrobe CRUD, outfit generation) and integration testing of workflows that cross components. Integration tests will verify that image upload works correctly, that the backend can forward an image to the FastAPI scanning service, and that predicted fields can be returned to the frontend and saved after user confirmation. The team will also perform end-to-end manual testing of the main user flows: sign up/login, add items manually, add items via scan, generate outfits with filters, save outfits, and preview outfits. Bugs found during testing will be tracked and prioritized to ensure demo-day stability.

12.0 Installation Instructions

To run the project locally, the user will install Node.js for the Express backend and React frontend, PostgreSQL for the database, and Python for the FastAPI scanning service. After cloning the repository, the user will configure environment variables (database connection string, JWT secret, and service URLs), install dependencies for both frontend and backend, and initialize the database schema. The backend server will be started first, followed by the frontend development server. If the ML scanning service is enabled, the user will also install Python dependencies and start the FastAPI server; otherwise, the application will still support manual clothing entry. A complete step-by-step setup guide will be included in the final submission README.

13.0 Future Implementation

Future improvements can expand both realism and personalization. Potential additions include improved background removal and segmentation for cleaner clothing extraction, more detailed attribute prediction (patterns, materials, brands), and personalized recommendations based on user feedback (liked/disliked outfits). The preview experience can be enhanced by supporting more mannequin customization and better alignment of clothing to the model. Additional features could include calendar-based outfit planning, weather-based suggestions, and optional sharing of outfits with friends.

13.1 Risks and Mitigations

The project includes several risks that will be managed through design choices. Clothing scanning accuracy may vary due to lighting and background noise; to mitigate this, scan results will be treated as suggestions and users can always edit fields before saving. The 3D preview may be more complex than expected; to mitigate this, the initial version will prioritize a stable mannequin preview and allow a simpler visualization method if needed. Deployment can also be challenging because the system uses both Node and Python services; to mitigate this, the team will provide clear installation steps and keep the core app functional even when the scanning service is disabled.

14.0 References

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